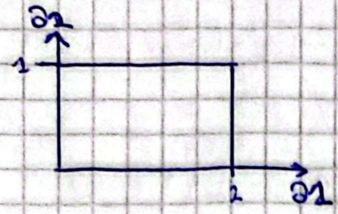
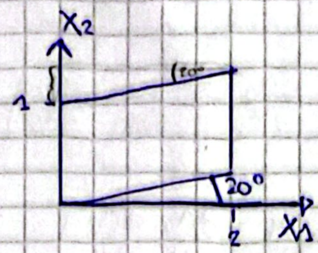
Recuperatorio Parcial 2 2024

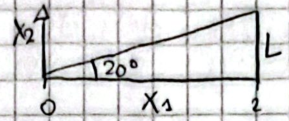
$Tg 20° = l$

1)





a) El campo de desplazamientos



X_2 se desvía 20° ¿cuánto es su longitud?

$$Tan 20° = \frac{L}{X_1} \rightarrow L = Tan 20° . X_1$$

Descripción de movimiento

$$\begin{cases} X_1 = a_1 \\ X_2 = \underbrace{a_2}_{\text{altura que gana}} + \underbrace{Tg 20° X_1}_{\text{altura ganada}} \rightarrow a_2 + Tg 20° a_1 \end{cases}$$

Campo de desplazamiento $U_i = X_i - a_i$

$$\begin{cases} U_1 = X_1 - a_1 \rightarrow a_1 - a_1 = 0 \\ U_2 = X_2 - a_2 \rightarrow (a_2 + Tg 20° a_1) - a_2 \rightarrow Tg 20° a_1 \end{cases}$$

b) El Tensor de deformaciones de Green Lagrange

$$\varepsilon_{ij} = \frac{1}{2}\left(\frac{\partial X_k}{\partial a_i} \cdot \frac{\partial X_k}{\partial a_j} - \delta_{ij}\right)$$

$\delta_{ij} = 1 \wedge i = j$

$\delta_{ij} = 0 \wedge i \neq j$

$$\varepsilon_{11} = \frac{1}{2}\left(\frac{\partial X_1}{\partial a_1} \cdot \frac{\partial X_1}{\partial a_1} + \frac{\partial X_2}{\partial a_1} \cdot \frac{\partial X_2}{\partial a_1} - \delta_{ij}\right)$$

$$\varepsilon_{11} = \frac{1}{2}\left(1 \cdot 1 + Tg 20° . Tg 20° - 1\right) = \frac{1}{2} Tg^2(20)$$

$$\varepsilon_{12} = \varepsilon_{21} = \frac{1}{2}\left(\frac{\partial X_1}{\partial a_1} \cdot \frac{\partial X_1}{\partial a_2} + \frac{\partial X_2}{\partial a_1} \frac{\partial X_2}{\partial a_2} - \delta_{12}\right)$$

$$\frac{1}{2}\left(1 \cdot 0 + Tg 20° . 1 - 0\right) = \frac{1}{2} Tg 20°$$

$$\varepsilon_{22} = \frac{1}{2}\left(\frac{\partial X_1}{\partial a_2} \cdot \frac{\partial X_1}{\partial a_2} + \frac{\partial X_2}{\partial a_2} \frac{\partial X_2}{\partial a_2} - \delta_{22}\right) =$$

$$\frac{1}{2}\left(0 \cdot 0 + 1 \cdot 1 - 1\right) = 0$$

$$\varepsilon_{ij} = \begin{bmatrix} 1/2\, Tg^2 20 & 1/2\, Tg 20° \\ \frac{1}{2} Tg 20 & 0 \end{bmatrix}$$

c) Tensor de deformaciones Almansi

$$\varepsilon_{ij} = \frac{1}{2}\left(\delta_{ij} - \frac{\partial a_k}{\partial X_i} \cdot \frac{\partial a_k}{\partial X_j}\right)$$

$$\varepsilon_{11} = \frac{1}{2}\left(\delta_{11} - \frac{\partial a_1}{\partial X_1}\frac{\partial a_1}{\partial X_1} - \frac{\partial a_2}{\partial X_1}\frac{\partial a_2}{\partial X_1}\right) = \frac{1}{2}\left(1 - 1.1 - (-Tg 20^\circ)(-Tg 20^\circ)\right) = -\frac{1}{2}$$

$$\varepsilon_{22} = \frac{1}{2}\left(\delta_{22} - \frac{\partial a_1}{\partial X_2}\frac{\partial a_1}{\partial X_2} - \frac{\partial a_2}{\partial X_2}\frac{\partial a_2}{\partial X_2}\right) = \frac{1}{2}\left(1 - 0.0 - 1.1\right) = 0$$

$$\varepsilon_{12} = \frac{1}{2}\left(\delta_{12} - \frac{\partial a_1}{\partial X_1}\frac{\partial a_1}{\partial X_2} \cdot \frac{\partial a_2}{\partial X_1}\frac{\partial a_2}{\partial X_2}\right) = \frac{1}{2} Tg 20$$

2 a) Verificar equilibrio

$$\theta_{xx} = \frac{3}{4}\frac{q}{c^3}\left[\left(L^2 - \frac{2}{5}c^2 - x^2\right)y + \frac{2}{3}y^3\right]$$

$$\theta_{yy} = -\frac{3}{4}\frac{q}{c^3}\left(\frac{1}{3}y^3 - c^2 y + \frac{2}{3}c^3\right)$$

$$\theta_{xy} = -\frac{3}{4}\frac{q}{c^3}(c^2 - y^2)x \qquad \theta_{xz} = \theta_{yz} = \theta_{zz} = 0$$

1 $$\frac{\partial\theta_{xx}}{\partial x} + \frac{\partial\theta_{xy}}{\partial y} + \frac{\partial\theta_{xz}}{\partial z} + b_1 = 0 \rightarrow \cancel{\frac{3}{4}\frac{q}{c^3}(-2xy)} - \cancel{\left(\frac{3}{4}\frac{q}{c^3}\cdot -2yx\right)} = 0 \checkmark$$

2 $$\frac{\partial\theta_{xy}}{\partial x} + \frac{\partial\theta_{yy}}{\partial y} + \frac{\partial\theta_{yz}}{\partial z} + b_2 = 0 \rightarrow -\frac{3}{4}\frac{q}{c^3}(c^2 - y^2) - \frac{3}{4}\frac{q}{c^3}\left(\frac{3}{3}y^2 - c^2\right) = 0 \checkmark$$

3 $$\frac{\partial\theta_{xz}}{\partial x} + \frac{\partial\theta_{yz}}{\partial y} + \frac{\partial\theta_{zz}}{\partial z} + b_3 = 0 \qquad 0 + 0 + 0 + 0 = 0 \checkmark$$

Equilibrio

b) Determinar componentes de deformación: $\theta_{ij} = \lambda\varepsilon_{kk}\delta_{ij} + 2\mu\varepsilon_{ij}$ 1 incógnita

$\hookrightarrow \varepsilon_{kk} = \varepsilon_{xx} + \varepsilon_{yy} + \varepsilon_{zz}$

$\theta_{ij} = \lambda\varepsilon_{kk}\delta_{ij} + 2\mu\varepsilon_{ij}$ Tomando $i,j = k$

$\theta_{kk} = \lambda\varepsilon_{kk}\delta_{kk} + 2\mu\varepsilon_{kk}$

$\theta_{kk} = \lambda\varepsilon_{kk}3 + 2\mu\varepsilon_{kk} \rightarrow \left[\varepsilon_{kk} = \frac{\theta_{kk}}{3\lambda + 2\mu}\right]$ reemplazo en 1

$\theta_{ij} = \lambda\left(\frac{\theta_{kk}}{3\lambda + 2\mu}\right)\delta_{ij} + 2\mu\varepsilon_{ij}$ Algebra

$$\left[\varepsilon_{ij} = \frac{1}{2\mu}\theta_{ij} - \frac{\lambda}{2\mu(3\lambda + 2\mu)}\theta_{kk}\delta_{ij}\right]$$ Con esto podemos calcular las 6 componentes

$$\varepsilon_{ij} = \frac{1}{2\mu}\sigma_{ij} - \frac{\lambda}{2\mu(3\lambda+2\mu)}\sigma_{kk}\delta_{ij} \qquad \text{llamos } C = \frac{\lambda}{2\mu(3\lambda+2\mu)}$$

$$\left[\varepsilon_{ij} = \frac{1}{2\mu}\sigma_{ij} - C(\sigma_{xx}+\sigma_{yy})\delta_{ij}\right]$$

$$\varepsilon_{xy} = \frac{1}{2\mu}\sigma_{xy} - C(\sigma_{xx}+\sigma_{yy})\overset{\to 0}{\delta_{xy}} \to \frac{1}{2}\mu\sigma_{xy} \to \frac{1}{2}\mu \cdot -\frac{3}{4}\frac{q}{c^3}(c^2-y^2)x$$

$$\varepsilon_{xz} = \frac{1}{2\mu}\sigma_{xz} \to \frac{1}{2}\mu \cdot 0 \to 0$$

$$\varepsilon_{yz} = \frac{1}{2\mu}\cdot\sigma_{yz} \to \frac{1}{2}\mu\cdot 0 \to 0$$

$$\varepsilon_{xx} = \frac{1}{2\mu}\sigma_{xx} - C\cdot(\sigma_{xx}+\sigma_{yy})$$

$$\varepsilon_{yy} = \frac{1}{2\mu}\sigma_{yy} - C\cdot(\sigma_{xx}+\sigma_{yy})$$

$$\varepsilon_{zz} = \frac{1}{2\mu}\overset{\to 0}{\sigma_{zz}} - C(\sigma_{xx}+\sigma_{yy}) \to -C(\sigma_{xx}+\sigma_{yy})$$

c) Verificar compatibilidad de deformaciones $\frac{\partial^2\varepsilon_{xx}}{\partial y^2} + \frac{\partial^2\varepsilon_{yy}}{\partial x^2} = 2\frac{\partial\varepsilon_{xy}}{\partial x\partial y}$

Primero derivamos σ_{xx} y σ_{yy} ya que ε_{xx} ε_{yy} y ε_{xy} depende de σ_{xx} y σ_{yy}

$k = \frac{3}{4}\frac{q}{c^3}$

$$\sigma_{xx} = k\left[\left(L^2 - \frac{2}{5}c^2 - x^2\right)y + \frac{2}{3}y^3\right] \qquad \sigma_{yy} = -k\left(\frac{1}{3}y^3 - c^2y + \frac{2}{3}c^3\right)$$

$$\frac{\partial\sigma_{xx}}{\partial y} = k\left[\left(L^2 - \frac{2}{5}c^2 - x^2\right) + 2y^2\right] \qquad \frac{\partial\sigma_{yy}}{\partial x} = 0$$

$$\left[\frac{\partial^2\sigma_{xx}}{\partial y^2} = k4y\right] \qquad \left[\frac{\partial^2\sigma_{yy}}{\partial x^2} = 0\right]$$

$$\frac{\partial\sigma_{xx}}{\partial x} = k\cdot -2xy \qquad \frac{\partial\sigma_{yy}}{\partial y} = -k(y^2-c^2)$$

$$\left[\frac{\partial^2\sigma_{xx}}{\partial x^2} = -2ky\right] \qquad \left[\frac{\partial^2\sigma_{yy}}{\partial y^2} = -k2y\right]$$

$\theta_{xy} = -K(c^2-y^2)x$

$$\frac{\partial \theta_{xy}}{\partial x} = -K(c^2-y^2)$$

$$\left[\frac{\partial^2 \theta_{xy}}{\partial x \partial y} = -K(-2y)\right]$$

Reemplazamos

$$\frac{\partial^2 \varepsilon_{xx}}{\partial y^2} + \frac{\partial \varepsilon_{yy}}{\partial x^2} = 2\frac{\partial^2 \varepsilon_{xy}}{\partial x \partial y}$$

Lado der:

1 $\frac{\partial^2}{\partial y^2}\left(\frac{1}{2u}\theta_{xx} - C(\theta_{xx}+\theta_{yy})\right) \rightarrow \frac{1}{2u}\left[\frac{\partial^2 \theta_{xx}}{\partial y^2} - C.\left(\frac{\partial \theta_{xx}}{\partial y^2} + \frac{\partial^2 \theta_{yy}}{\partial y^2}\right)\right]$

$\frac{1}{2}u\ k4y - C.[k4y + -k2y] \rightarrow \frac{1}{2}u[k4y] - C[2ky] \rightarrow \frac{4ky}{2u} - C2ky$

2 $\frac{\partial^2}{\partial x_2}\left(\frac{1}{2u}\theta_{yy} - C(\theta_{xx}+\theta_{yy})\right) \rightarrow \frac{1}{2u}\left[\frac{\partial \theta_{yy}}{\partial x^2}\right] - C\left(\frac{\partial \theta_{xx}}{\partial x^2} + \frac{\partial \theta_{yy}}{\partial x^2}\right)$

$\frac{1}{2}u.[0] - C.(-2ky + 0] \rightarrow C2ky$

1 + 2 $= \frac{4ky}{2u} - \cancel{C2ky} + \cancel{C2ky} = \frac{2ky}{u}$

Lado Izq:

$2.\frac{\partial^2 \varepsilon_{xy}}{\partial x \partial y} = 2.\frac{1}{2u}\left(\frac{\partial^2 \theta_{xy}}{\partial x \partial y}\right) \rightarrow \frac{1}{u}.2ky \rightarrow \frac{2ky}{u}$

1 + 2 = 3

$\frac{2ky}{u} = \frac{2ky}{u}$ ✓

3) $V_x = k\frac{\partial\theta}{\partial y}$ $V_y = -k\frac{\partial\theta}{\partial x}$ $V_z=0$ $\theta(x,y)$

irrotacional $\nabla\times V=0$

definición de irrotacional $\nabla\times V = \left[\begin{matrix} e_x & e_y & e_z \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ V_x & V_y & V_z \end{matrix}\right|\begin{matrix} e_x & e_y \\ \partial/\partial x & \partial/\partial y \\ V_x & V_y \end{matrix}\Bigg|$

$$\left[\left(\cancel{e_x\frac{\partial}{\partial y}\overset{0}{V_z}}\right)+\left(e_y\frac{\partial}{\partial z}V_x\right)+\left(e_z\frac{\partial}{\partial x}V_y\right)\right]-\left[\left(V_x\frac{\partial}{\partial y}e_z\right)+\left(V_y\frac{\partial}{\partial z}e_x\right)+\left(\cancel{V_z\frac{\partial}{\partial x}\overset{0}{e_y}}\right)\right]$$

$$\left(e_y\frac{\partial}{\partial z}V_x\right)+\left(e_z\frac{\partial}{\partial x}V_y\right)-\left(V_x\frac{\partial}{\partial y}e_z\right)-\left(V_y\frac{\partial}{\partial z}e_x\right)$$

$$e_x\left(-\frac{\partial}{\partial z}V_y\right)+e_y\left(\frac{\partial V_x}{\partial z}\right)+e_z\left(\frac{\partial V_y}{\partial x}-\frac{\partial V_x}{\partial y}\right)$$

$$e_x\underbrace{\left(\frac{-\partial}{\partial z}\left(\frac{-k\partial\theta(x,y)}{\partial x}\right)\right)}_{0}+e_y\underbrace{\left(\frac{\partial}{\partial z}\left(\frac{k\partial\theta(x,y)}{\partial y}\right)\right)}_{0}+e_z\left(\frac{\partial}{\partial x}\left(\frac{-k\partial\theta(x,y)}{\partial x}\right)-\frac{\partial}{\partial y}\frac{k\partial\theta}{\partial y}\right)$$

porque θ no depende de z

$$e_z\left(\frac{\partial}{\partial x}\frac{\partial}{\partial x}\theta\cdot -k-\frac{\partial}{\partial y}\frac{\partial}{\partial y}\theta\cdot k\right)$$

$$e_z\left(-k\frac{\partial^2\theta}{\partial x^2}-k\frac{\partial^2\theta}{\partial y^2}\right)$$ No es irrotacional ya que $\neq 0$

Incompresible $\text{div}(V)=0 \rightarrow \nabla\cdot V$
divergencia

$$\nabla\cdot V=\frac{\partial V_x}{\partial x}+\frac{\partial V_y}{\partial y}+\frac{\partial V_z}{\partial z}\rightarrow\frac{\partial}{\partial x}\frac{k\partial\theta}{\partial y}-\frac{\partial}{\partial y}\frac{k\partial\theta}{\partial x}+0$$

$$\cancel{k\frac{\partial}{\partial x}\frac{\partial\theta}{\partial y}}-\cancel{k\frac{\partial}{\partial y}\frac{\partial\theta}{\partial x}}=0$$ es incompresible

Tasa de deformación: $V_{ij} = \frac{1}{2}\left(\frac{\partial V_i}{\partial x_j} + \frac{\partial V_j}{\partial x_i}\right)$

$$V_{xx} = \frac{1}{2}\left(\frac{\partial V_x}{\partial x} + \frac{\partial V_x}{\partial x}\right) = \frac{1}{2}\left(2\frac{\partial V_x}{\partial x}\right) = \frac{\partial V_x}{\partial x} \longrightarrow \frac{\partial V_x}{\partial x} = \frac{k\,\partial^2\theta}{\partial x \partial y}$$

$$V_{yy} = \frac{\partial V_y}{\partial y} = -\frac{k\,\partial^2\theta}{\partial x \partial y}$$

$$V_{zz} = \frac{\partial V_z}{\partial z} = 0$$

$$V_{xy} = \frac{1}{2}\left(\frac{\partial V_x}{\partial y} + \frac{\partial V_y}{\partial x}\right) = \frac{1}{2}\left(\frac{k\,\partial\theta}{\partial y^2} - \frac{k\,\partial\theta}{\partial x^2}\right) = \frac{k}{2}\left(\frac{\partial\theta}{\partial y^2} - \frac{\partial\theta}{\partial x^2}\right)$$

$$V_{xz} = \frac{1}{2}\left(\frac{\partial V_x^{\,0}}{\partial z} + \frac{\partial V_z^{\,0}}{\partial x}\right) = 0$$

$$V_{yz} = \frac{1}{2}\left(\frac{\partial V_y}{\partial z} + \frac{\partial V_z}{\partial y}\right) = 0$$

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Ej 4) Partiendo de la ec. del balance de cantidad de movimiento angular demostrar que el Tensor de Tensiones debe ser simétrico considerando la ec. de movimiento y conservación de masa

Cant de mov angular: $H_i = \int_V \varepsilon_{ijk} X_j \rho V_k \, dV$ (1)

Momento de fuerzas externas $L_i = \int_V \varepsilon_{ijk} X_j b_k \, dV + \int_S \varepsilon_{ijk} X_j T_k \, dS$ (2)

Balance: $\frac{DH_i}{Dt} = L_i$ (3)

(1) y (2) en (3)

$$\frac{D}{Dt}\left(\int_V \varepsilon_{ijk} X_j \rho V_k \, dV\right) = \int_V \varepsilon_{ijk} X_j b_k \, dV + \int \varepsilon_{ijk} X_j T_k \, dS$$ metemos la derivada por conservación de masa

$$* \quad \int_V \rho \frac{D}{Dt}(\varepsilon_{ijk} X_j V_k) \, dV = \int_V \varepsilon_{ijk} X_j b_k \, dV + \int_S \varepsilon_{ijk} X_j T_k \, dS$$ aplicamos regla del producto

$$\varepsilon_{ijk}\left[\left(\frac{D}{Dt}X_j\right)V_k + X_j\left(\frac{D V_k}{Dt}\right)\right] \quad \left[\frac{DX_j}{Dt} = V_j\right.$$ derivada de la pos respecto al tiempo

$$\varepsilon_{ijk} V_j V_k + \varepsilon_{ijk} X_j \left(\frac{D}{Dt} V_k\right) \rightarrow$$

$$V \times V + \varepsilon_{ijk} X_j \left(\frac{D}{Dt} V_k\right) \rightarrow \qquad V \times V = 0$$

volvemos a *

$$\int_V \rho \left(\frac{D}{Dt} V_k\right) \varepsilon_{ijk} X_j \, dV = \int_V \varepsilon_{ijk} X_j b_k \, dV + \int_S \varepsilon_{ijk} X_j T_k \, dS$$

Usamos Cauchy $T_i = \sigma_{ji} n_j \rightarrow T_k = n_\ell \sigma_{\ell k}$ en el último término

$$\int_V \rho \left(\frac{D}{Dt} V_k\right) \varepsilon_{ijk} X_j \, dV = \int_V \varepsilon_{ijk} X_j b_k \, dV + \int_S \varepsilon_{ijk} X_j \sigma_{\ell k} n_\ell \, dS$$

Aplicamos Gauss al ultimo Termino $\int_S A_i n_i \, ds = \int_V \frac{\partial A_i}{\partial x_i} dV$

$$\int_V \rho\left(\frac{D}{Dt}V_k\right)\varepsilon_{ijk}X_j \, dV = \int_V \varepsilon_{ijk}X_j b_k \, dV + \int_V \frac{\partial}{\partial x_l}(\varepsilon_{ijk}X_j\sigma_{lk})\, dV$$ aplicamos regla del producto

$$\int_V \rho\left(\frac{D}{Dt}V_k\right)\varepsilon_{ijk}X_j \, dV = \int_V \varepsilon_{ijk}X_j b_k \, dV + \int_V \varepsilon_{ijk}\left[\left(\frac{\partial X_j}{\partial x_l}\right)\sigma_{lk} + X_j\left(\frac{\partial \sigma_{lk}}{\partial x_l}\right)\right] dV$$ separando el Termino

$\rightarrow \delta_{jl}\sigma_{lk} \rightarrow \sigma_{jk}$

$$\int_V \rho\left(\frac{D V_k}{Dt}\right)\varepsilon_{ijk}X_j \, dV = \int_V \varepsilon_{ijk}X_j b_k \, dV + \int_V \varepsilon_{ijk}\sigma_{jk}\, dV + \int_V \varepsilon_{ijk}X_j \frac{\partial \sigma_{lk}}{\partial x_l} dV$$

$$\int_V \varepsilon_{ijk}X_j \, \rho\left(\frac{D V_k}{Dt}\right) dV - \int_V \varepsilon_{ijk}X_j b_k \, dV - \int_V \varepsilon_{ijk}\sigma_{jk}\, dV - \int \varepsilon_{ijk}X_j \frac{\partial \sigma_{lk}}{\partial x_l} dV = 0$$

$$\int_V \varepsilon_{ijk}X_j\left[\rho\frac{DV_k}{Dt} - b_k - \frac{\partial \sigma_{lk}}{\partial x_l}\right] dV - \int \varepsilon_{ijk}\sigma_{jk}\, dV = 0$$

$\rightarrow$ Ec de movimiento $\rho\frac{DV_i}{Dt} = \frac{\partial \sigma_{ij}}{\partial x_j} + b_i \rightarrow \left[\rho\frac{DV_i}{Dt} - b_i - \frac{\partial \sigma_{ij}}{\partial x_j} = 0\right]$

$$\int_V \varepsilon_{ijk}X_j[0]\, dV - \int_V \varepsilon_{ijk}\sigma_{jk}\, dV = 0$$

$$-\int_V \varepsilon_{ijk}\sigma_{jk}\, dV = 0$$ Para que esta integral de 0, lo que esta adentro debe dar 0 $\varepsilon_{ijk}\sigma_{jk} = 0$

$\varepsilon_{ijk}\sigma_{jk} = 0$ Tomamos $i=1$ y j y $k = 2,3$ no se pueden repetir

$\varepsilon_{123}\sigma_{23} + \varepsilon_{132}\sigma_{32}$ $\quad \varepsilon_{132} = -\varepsilon_{132}$ por permutación de indices

$$\varepsilon_{123}\sigma_{23} - \varepsilon_{132}\sigma_{32} = 0$$

$$\cancel{\varepsilon_{123}}\sigma_{23} = \cancel{\varepsilon_{123}}\sigma_{32}$$

$$\sigma_{23} = \sigma_{32}$$ ✓ Demostramos que debe ser simetrico

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5) $X_1 = a_1 e^{-t}$ $\quad X_2 = a_2 e^{t}$ $\quad X_3 = a_3 + a_2(e^{-t} - 1)$

El componente vectorial A_i está dado por

$$A_1(X_1, X_2, X_3, t) = e^{-t}(X_1 - 2X_2 + 3X_3)$$
$$A_2(X_1, X_2, X_3, t) = e^{-2t}(X_1 - 2X_2 + 3X_3)$$
$$A_3(X_1, X_2, X_3, t) = e^{-3t}(X_1 - 2X_2 + 3X_3)$$

Calcule la derivada material A_i en ese medio con el resultado final en descripción euleriana

Formula Derivada material: $$\frac{DA_i}{Dt} = \underbrace{\frac{\partial A_i}{\partial t}}_{\text{cambio local}} + \underbrace{\frac{\partial A_i}{\partial X_j} V_j}_{\text{cambio convectivo}}$$

Paso 1: Encontrar el campo de velocidades V_i $\qquad V = \left(\frac{\partial X_1}{\partial t} ; \frac{\partial X_2}{\partial t} ; \frac{\partial X_3}{\partial t}\right)$

Dirección 1:

$X_1 = a_1 e^{-t} \rightarrow a_1 = X_1 . e^{t}$

derivamos $V_1: \frac{\partial X_1}{\partial t} = -a_1 e^{-t}$

Euleriano: V en función de las X

Reemplazamos: $V_1 = -(X_1 e^{t}) e^{-t} \rightarrow -X_1$

Dirección 2:

$X_2 = a_2 e^{t} \qquad a_2 = X_2 e^{-t}$

$V_2 = \frac{\partial X_2}{\partial t} = a_2 e^{t}$

reemplazamos $V_2: X_2 e^{-t} . e^{t} \rightarrow X_2$

Dirección 3

$X_3 = a_3 + a_2(e^{-t} - 1) \rightarrow$

$V_3 = \frac{\partial X_3}{\partial t} = -a_2 e^{-t}$

reemplazamos $V_3 = -X_2 e^{-t} e^{-t} = -X_2 e^{-2t}$

$$V = (-X_1 ; X_2 ; -X_2 e^{-2t})$$

local   Convectiva

Paso2: Calcular derivadas

$$\frac{DA}{Dt} = \frac{\partial A}{\partial t} + \left[V_1 \frac{\partial A}{\partial x_1} + V_2 \frac{\partial A}{\partial x_2} + V_3 \frac{\partial A}{\partial x_3} \right]$$

$$\frac{\partial A_1}{\partial t} = -e^{-t}(x_1 - 2x_2 + 3x_3) \qquad V_2 \frac{\partial A_1}{\partial x_2} = X_2 . e^{-t} . -2$$

$$V_1 \frac{\partial A_1}{\partial x_1} = -X_1 . + e^{-t} . 1 \qquad V_3 \frac{\partial A_1}{\partial x_3} = -X_2 e^{-2t} . 3 . e^{-t}$$

Sumamos $\frac{\partial A_1}{\partial t} + V_j \frac{\partial A_1}{\partial x_j}$

$$\frac{DA_1}{Dt} = -e^{-t}(X_1 - 2X_2 + 3X_3) + \left(-X_1 e^{-t} + -2X_2 e^{-t} - 3X_2 e^{-3t} \right)$$

$$\left[\frac{DA_1}{Dt} = -2X_1 e^{-t} - 3X_3 e^{-t} - 3X_2 e^{-3t} \right]$$

$$\frac{\partial A_2}{\partial t} = -2e^{-2t}(X_1 - 2X_2 + 3X_3) \qquad V_2 \frac{\partial A_2}{\partial x_2} = X_2 . -+e^{-2t} . (-2)$$

$$V_1 \frac{\partial A_2}{\partial x_1} = -X_1 . + e^{-2t}(1) \qquad V_3 \frac{\partial A_2}{\partial x_3} = -X_2 e^{-2t} . -+e^{2t} . 3$$

$$\frac{DA_2}{\partial t} = -2e^{-2t}(X_1 - 2X_2 + 3X_3) + -X_1 e^{-2t} - 2X_2 e^{-2t} - 3X_2 e^{-4t}$$

$$\left[\frac{DA_2}{\partial t} = -3X_1 e^{-2t} + 2X_2 e^{-2t} - 6X_3 e^{-2t} - 3X_2 e^{-4t} \right]$$

$$\frac{\partial A_3}{\partial t} = -3e^{-3t}(X_1 - 2X_2 + 3X_3) \qquad V_2 \frac{\partial A_3}{\partial x_2} = X_2 . e^{-3t} . -2$$

$$V_1 \frac{\partial A_3}{\partial x_1} = -X_1 e^{-3t} \qquad V_3 \frac{\partial A_3}{\partial x_3} = -X_2 e^{-t} . e^{-3t} . 3$$

$$\frac{DA_3}{\partial t} = -3e^{-3t}(X_1 - 2X_2 + 3X_3) + -X_1 e^{-3t} - 2X_2 e^{-3t} - 3X_2 e^{-5t}$$

$$\left[\frac{DA_3}{\partial t} = -4X_1 e^{-3t} + 4X_2 e^{-3t} - 9X_3 e^{-3t} - 3X_2 e^{-5t} \right]$$

Aceleración en descripción Euleriana

$$a_i = \frac{\partial v_i}{\partial t} + v_j \frac{\partial v_i}{\partial x_j}$$